Regime-Based Asset Allocation using Markov Chains and Hidden Markov Models.

By
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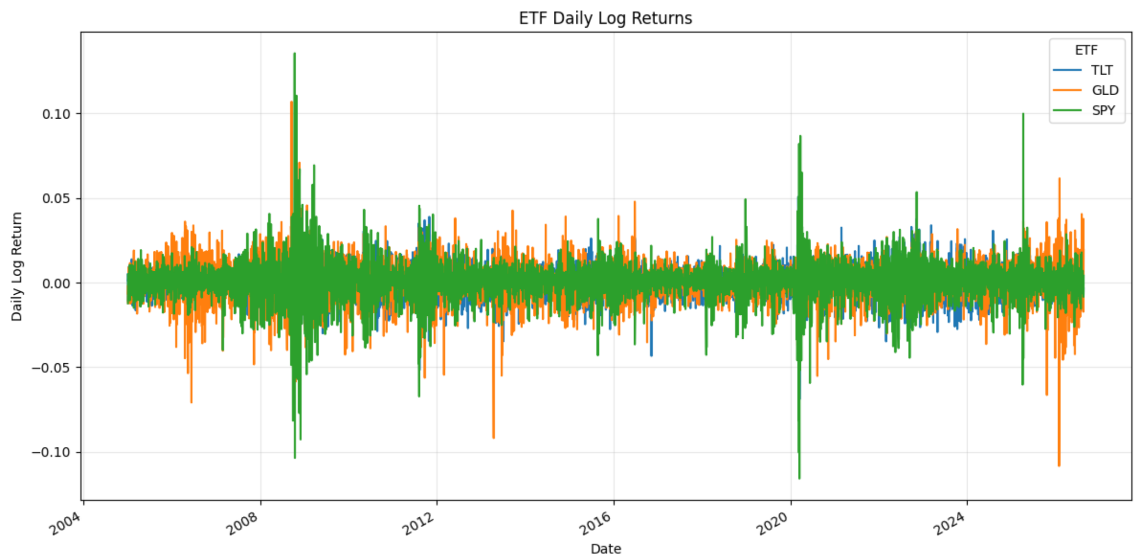
Step 1: Data Preparation & Exploration

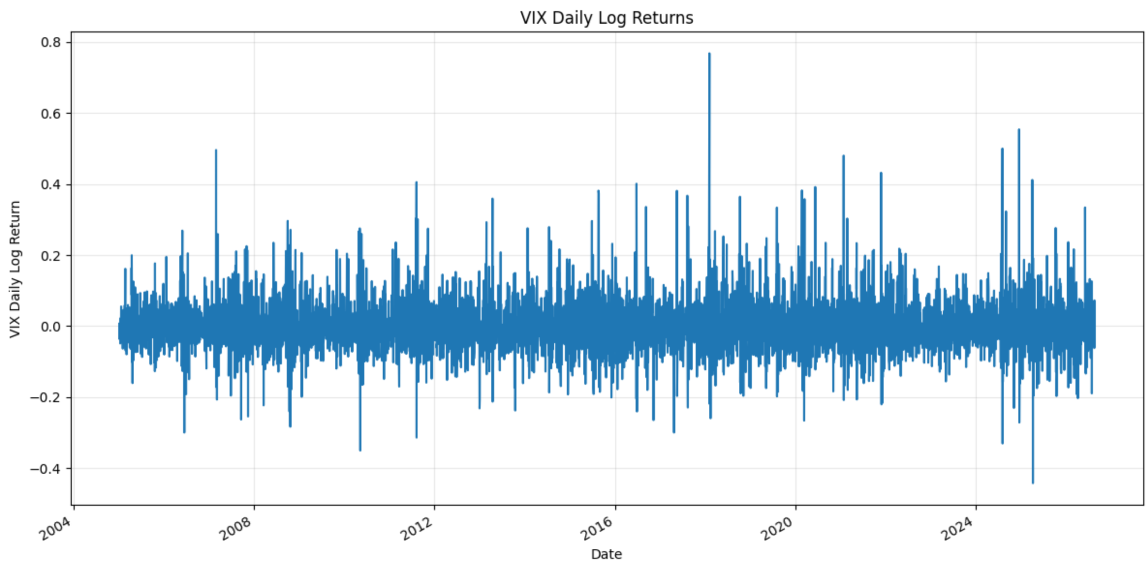
We downloaded the daily price data from yahoo Finance for TLT (Treasury ETF), GLD (Gold ETF), SPY (S&P 500 ETF) and VIX (Volatility Index) and the system automatically aligned all series to the maximum common sample period from January 4, 2005 to August 18, 2026 which ensures that every date in the dataset has values for all four instruments (Yahoo Finance).

The earliest common date corresponds to the inception of GLD (the gold ETF) because it began trading in late 2004 thus the dataset starts from 2005 to maintain consistency.



Graph 1: ETF Daily Log Returns

The graph above illustrates how equities (SPY), bonds (TLT) and gold (GLD) respond differently to market conditions. SPY shows sharp swings during crises like 2008 and 2020, TLT often rises when equities fall as investors seek safety in U.S. Treasuries and GLD remain resilient as a hedge against uncertainty. Together, the overlapping lines highlight contrasting asset behaviors and provide insight into regime-based allocation strategies.



Graph 2: VIX Daily Log Returns

The VIX returns graph shows how market volatility changes over time. Sharp spikes appear during stress events such as 2008, 2016 and 2020, reflecting sudden surges in uncertainty. Volatility is mean-reverting with rapid jumps followed by gradual declines. These dynamics make the VIX a reliable indicator for defining regimes, which can guide asset rotation strategies (Cboe Global Markets).

Step 2: Modeling VIX Regimes

Here we classify volatility regimes in the VIX series.

Discrete Markov Chain Approach

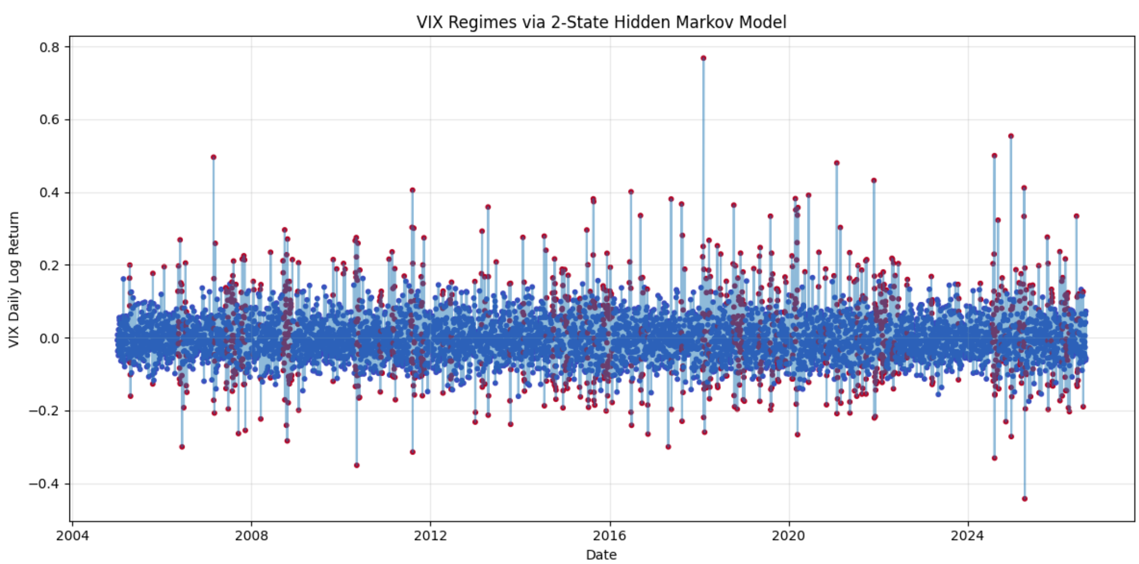
VIX log returns were discretized into three regimes: low, medium and high volatility, and the transition matrix below shows the probability of moving from one regime to another.

Transition Matrix			
VIX	**0**	**1**	**2**
0	0.322563	0.328134	0.349304
1	0.271309	0.368802	0.359889
2	0.394595	0.294054	0.311351

Table 1: Transition matrix

The 33rd and 66th percentile thresholds were used to separate VIX daily log returns into three regimes. State 0 (low VIX-return regime) was allocated to data below the 33rd percentile, State 1 (medium VIX-return regime) to observations between the 33rd and 66th percentiles, and State 2 (high VIX-return regime) to observations above the 66th percentile. In addition to producing roughly balanced groups, this quantile-based categorization offers the discrete states needed to estimate the transition matrix.

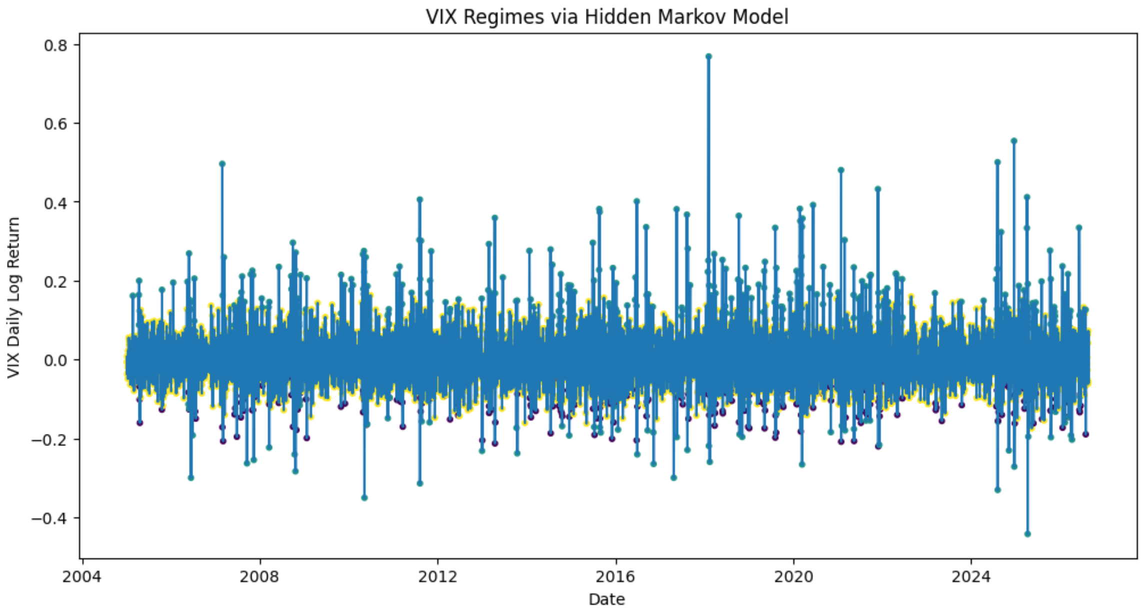
Hidden Markov Model Approach



Graph 3: VIX Regimes via 2-State Hidden Markov Model

The VIX daily log-return series was fitted to a two-state Gaussian Hidden Markov Model (HMM). An unobserved state in an HMM generates the observed return, and the state itself goes through a Markov transition process. The most-likely state sequence allocates each date to the state with the highest fitted probability, whereas posterior (smoothed) state probabilities can be calculated for every observation. This aligns with Rabiner's conventional HMM framework (257-86).

The VIX-return process is parsimoniously divided into two recurrent statistical contexts by the two-state definition. State numbers should not be automatically associated with specific economic meanings because they are arbitrary designations.



Graph 4: VIX Regimes via 3-State Hidden Markov Model

A smoother regime change than the discrete quantile technique was revealed when we applied a 3-state Hidden Markov Model (HMM) on VIX returns. Medium regimes serve as a bridge between calmer low-volatility states and high-volatility eras like 2008 and 2020.

The fitted hidden states provide a probabilistic way to identify recurrent volatility scenarios, and the observed VIX return is viewed as an emission generated by an unobserved state. This is consistent with the traditional HMM paradigm, which probabilistically connects latent Markov states to observable signals (Rabiner 257-86). In general, the HMM outperforms the Markov Chain method in capturing volatility persistence and mean-reversion.

Step 3: Model Comparison, State Selection and ETF Return Analysis.

Model Comparison

The two HMM options were compared on the same VIX-return sample using the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), and log-likelihood. A greater log-likelihood indicates a better fit to the observed data. AIC compares fit to additional model parameters (Akaike 716-23), whereas BIC adds a dependent sample size penalty for complexity (Schwarz 461-64). For AIC and BIC, lower values are preferred.

In order to guarantee consistency, the updated comparison uses the information-criterion parameter counts from the HMM library for both requirements. The discrete quantile Markov chain is retained as an interpretable benchmark instead of being direct equivalent under the Gaussian-HMM informational requirement.

Model	Log-Likelihood	AIC	BIC	Decision
2-State HMM	6959.92	-13905.83	-13859.62	Not selected
3-State HMM	7030.65	-14033.29	-13940.87	Selected

Table 2: Statistical Comparison of Two-State and Three-State HMMs

We used statistical criteria to compare option models (Markov Chain vs. HMM, 2 vs. 3 states).

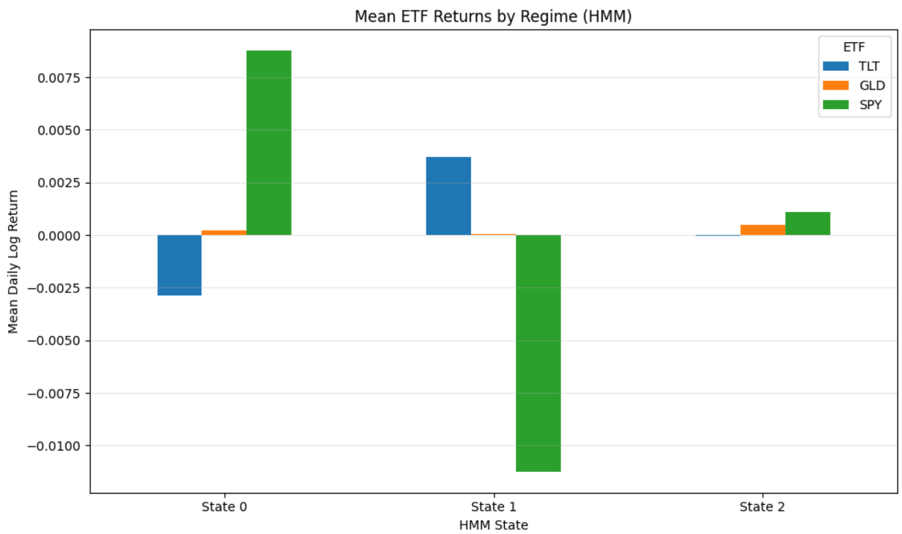
The output values show that the 3-state Hidden Markov Model has an excellent statistical match to the VIX data. The model provides a good explanation for the observed volatility patterns, as indicated by the log-likelihood of 7030.64. The model strikes a great balance between accuracy and complexity, as evidenced by the AIC of −14031.29 and BIC of −13932.26.

These findings demonstrate that the 3-state HMM is the ideal regime model for further ETF return study since it captures volatility dynamics better than both the discrete Markov Chain or a more simplified 2-state HMM.

ETF Returns by HMM State

HMM State	TLT Mean	TLT Std. Deviation	GLD Mean	GLD Std. Deviation	SPY Mean	SPY Std. Deviation	Highest Mean
0	-0.002892	0.010032	0.000241	0.012569	0.008754	0.012165	SPY
1	0.003699	0.011738	0.000049	0.016436	-0.011237	0.021277	TLT
2	-0.000048	0.008571	0.000483	0.010654	0.001072	0.008943	SPY

Table 3. State-Conditional Mean and Standard Deviation of Daily ETF Log Returns



Graph 5: Mean ETF Returns by Regime (3-State HMM)

In the fitted sample, State 0 is significantly equity-favoring: SPY has the highest mean daily log return at 0.008754, whereas GLD and TLT have mean daily log returns of 0.000241 and -0.002892, respectively. State 1 is defensive: SPY has the highest SPY standard deviation (0.021277) and a highly negative mean (-0.011237), while TLT has the highest mean (0.003699). Once more, SPY has the highest mean return in State 2 at 0.001072, surpassing GLD at 0.000483 and TLT at -0.000048.

Since they offer the empirical state-to-asset mapping needed in Step 4, these discrepancies have economic significance for the project. The portfolio choice is based on observed ETF performance inside each fitted state; the state numbers themselves remain labels.

Step 4: Designing the Rotation Strategy

Based on the latest executed state-conditional means, the correct allocation mapping is:

HMM State	Highest Mean ETF	Mean Daily Log Return	Target Allocation
State 0	SPY	0.008754	100% SPY
State 1	TLT	0.003699	100% TLT
State 2	SPY	0.001072	100% SPY

Table 4. Final State-to-Allocation Mapping

Since GLD does not have the highest mean return in any fitted state, the updated rule is that State 0 is SPY, State 1 is TLT, and State 2 is SPY. This only indicates that GLD does not succeed under the assignment's stringent highest-mean condition for this sample, not that gold is economically insignificant.

An execution lag of one trading day is used in the backtest. The asset held on day t+1 is determined by the state inferred on day t. By using the same day's state and return at the same time, this lag prevents a direct look-ahead bias.

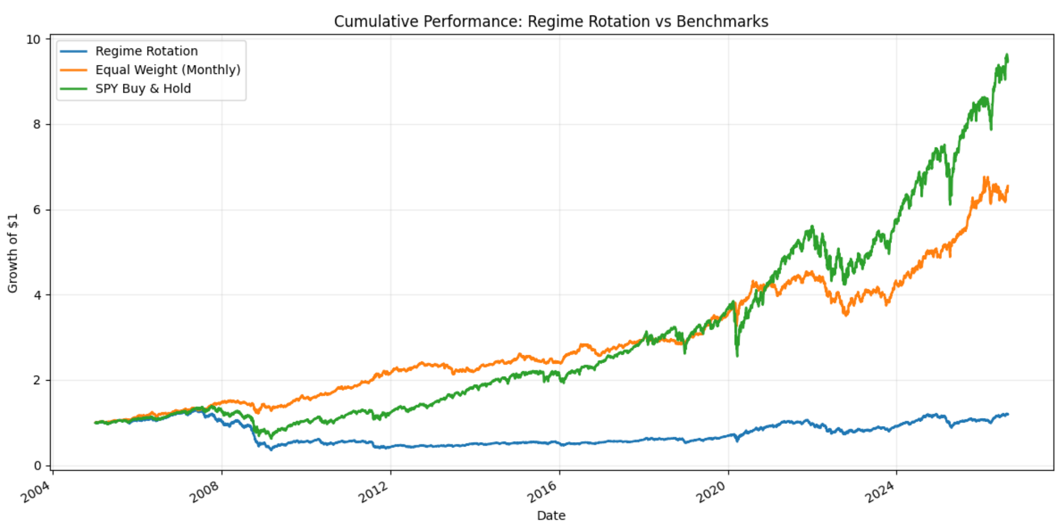
Step 5: Backtesting and Evaluation

ETF log returns are converted to simple returns prior to portfolio compounding, and the regime-strategy return using the one-day-lagged state signal is compared with two necessary benchmarks: (1) an equal-weight portfolio holding one-third TLT, one-third GLD, and one-third SPY with monthly rebalancing, and (2) buy-and-hold SPY.

Cumulative return, annualized return, annualized volatility, Sharpe ratio, and maximum drawdown are the necessary performance metrics. We assume 252 trading days annually and a 0% risk-free rate.

Metric	Definition Used in the Notebook
Cumulative Return	Ending wealth minus 1
Annualized Return	Geometric annualized growth rate using 252 trading days
Annualized Volatility	Daily standard deviation × √252
Sharpe Ratio	Daily mean / daily standard deviation × √252; risk-free rate = 0%
Maximum Drawdown	Largest peak-to-trough percentage decline in cumulative wealth

Table 5. Required Backtest Metrics



Graph 6: Cumulative Performance: Regime Rotation vs Benchmarks

Portfolio	Cumulative Return	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown
Regime Rotation	18.95%	0.81%	16.90%	0.132	-73.03%
Equal Weight (Monthly)	554.76%	9.09%	9.74%	0.943	-23.04%
SPY Buy & Hold	848.67%	10.98%	18.92%	0.646	-55.19%

Table 6. Performance Summary

Interpretation

The regime-based rotation approach significantly underperformed both benchmark portfolios over the sample period, according to the backtest. The method produced an annualized return of 0.81% and a cumulative return of just 18.95%. By contrast, the SPY buy-and-hold yielded the highest cumulative return of 848.67% and an annualized return of 10.98%, while the monthly equal-weight portfolio produced a cumulative return of 554.76% and an annualized return of 9.09%.

When analyzed on a risk-adjusted basis, the results are especially poor. Although the regime strategy yielded an annualized volatility of 16.90%, its Sharpe ratio was only 0.132. This indicates that each unit of risk taken yielded a comparatively low return. In comparison, the equal-weight portfolio had the lowest annualized volatility (only 9.74%), but it also had the highest Sharpe ratio (0.943). SPY had an annualized volatility of 18.92% and a Sharpe ratio of 0.646**.**

Maximum drawdown offers more proof of the regime strategy's shortcomings. The rotation portfolio's highest drawdown of roughly 73.03% was significantly more than the equal-weight portfolio's 23.04% fall and even greater than SPY's 55.19% drawdown. As a result, the regime strategy did not offer the downside protection that one might anticipate from alternating between defensive and equities assets.

Cumulative Performance

These conclusions are supported by the cumulative-performance graph. The most long-term wealth gain was achieved with SPY buy-and-hold, especially during the prolonged equity-market expansion that followed the global financial crisis. Throughout the period, the equal-weight portfolio similarly showed robust and comparatively steady growth.

The regime-rotation portfolio, on the other hand, concluded only marginally above its starting value and stayed relatively flat throughout most of the sample. Throughout the entire investment horizon, the strategy's gains were insufficient to reach either benchmark, regardless of occasional intervals of positive development.

Investment Implication

The findings show that a successful asset-allocation strategy does not always follow from the identification of statistically separate volatility regimes. Allocating 100% of the portfolio to the historically best-performing ETF within each state did not produce superior out-of-sample-like backtest performance when the one-day execution lag was applied, despite the fact that the three HMM states displayed significant differences in the historical average returns of SPY, TLT, and GLD.

Regime classifications may be subject to frequent changes or may only be recognized after market conditions have shifted. Therefore, the value of responding to the recognized regime may be diminished by the one-day execution delay. Furthermore, concentrated 100% allocations result from solely depending on the highest historical mean return, which ignores volatility, correlation, transaction costs, and uncertainty surrounding state identification.

Benchmark Comparison and Recommendation

The monthly equal-weight portfolio outperformed the other two in terms of overall risk-adjusted return. SPY saw far higher volatility and drawdown even if it produced the highest absolute cumulative and annualized returns. The equal-weight portfolio had the highest Sharpe ratio of 0.943, the lowest volatility of 9.74%, the smallest maximum drawdown of 23.04%, and a robust annualized return of 9.09%.

Therefore, the straightforward monthly equal-weight portfolio would be recommended for an investor looking for the best balance between return and risk based on the study's historical findings. An investor who is willing to tolerate higher volatility and drawdowns and is primarily looking for the highest long-term return would favor SPY buy-and-hold. Since the tested regime-rotation approach yielded far lower returns and worse downside-risk characteristics than either benchmark, it would not be advised in its current form.

Sensitivity to the Number of States

Strategy	Cumulative Return	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown
2-State Rotation	8416.55%	22.86%	15.24%	1.427	-30.42%
3-State Rotation	18.95%	0.81%	16.90%	0.132	-73.03%

Table 7: Sensitivity Analysis - 2-State vs. 3-State Regime Rotation Strategy

The number of states utilized has a significant impact on the results. With a total return of 8416.55%, an annualized return of 22.86%, and a Sharpe ratio of 1.427, the 2-state rotation approach significantly beat the 3-state method, which produced returns of 18.95%, 0.81%, and 0.132, respectively. Additionally, the maximum drawdown for the 2-state model was lower at -30.42% compared to -73.03% for the 3-state model. These findings imply that incorporating a third regime may add needless complexity and may not enhance the investing approach. As a result, it is evident that the number of states affects the strategy's performance; throughout the examined sample period, the 2-state specification produced much superior risk-adjusted performance.

Overall Objective Evaluation

Evaluating if regime-based rotation enhances risk-adjusted performance in comparison to straightforward benchmarks is the project's goal. The answer is no for the statistically chosen three-state HMM: the strategy's drawdown is significantly worse and its Sharpe ratio (0.132) is lower than both benchmark Sharpe ratios. Among the main comparative portfolios, the equal-weight portfolio is the best option in terms of risk adjustment.

However, under a different regime specification, regime-based rotation can appear significantly stronger, as seen by the two-state sensitivity result. Therefore, the proper conclusion is not that regime rotation is always unproductive, but rather that its effectiveness is extremely dependent on state specification and should be verified using tougher rolling-window or out-of-sample tests before being regarded as an appropriate investment strategy.

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